班级：_____

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主观题

					15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	0.5

1.（15分）已知圆 C 经过点 $A(2, 0)$，$B(0, 4)$，且圆心在直线 $x - y = 0$ 上。

(1) 求圆 C 的方程；(2) 求过点 $P(1, -1)$ 的圆 C 的切线方程。

解：(1) 设 $(x-a)^2+(y-a)^2=r^2$.

$$\therefore \begin{cases} (2-a)^2+a^2=r^2 \\ a^2+(4-a)^2=r^2 \end{cases}$$

$$\therefore \begin{cases} a=3 \\ r=\sqrt{10} \end{cases}$$

$\therefore (x-3)^2+(y-3)^2=10$

(2) 1° 斜率不存在.

$\therefore x=1$

$\therefore d=2\neq\sqrt{10}$，舍去

2° 斜率存在

设 $y+1=k(x-1)$

$y=kx-k-1$

$$d=\frac{|2k-1-3|}{\sqrt{k^2+1}}=\sqrt{10}.$$

$\therefore (2k-4)^2=10(k^2+1)$

$\therefore 3k^2+8k-3=0.$

$\therefore k=-3$ 或 $\frac{1}{3}$

$\therefore y=-3x+2$ 或 $y=\frac{1}{3}x-\frac{4}{3}$

$4k^2-16k+16=10k^2+10.$ $\frac{-8\pm10}{6}$ $\frac{-4\pm5}{3}$

$6k^2+16k-6=0$

2. （15分） 已知抛物线 C： $y^2 = 2px(p > 0)$ 的焦点为 F，过 F 的直线 l 与 C 交于 A， B 两点，且 $|AB| = 8$，线段 AB 的中点的横坐标为 3。

(1) 求抛物线 C 的方程；

(2) 求直线 l 的方程。

解：(1) 设 $A(\frac{u^2}{2p}, u)$

$B(\frac{v^2}{2p}, v)$

$\therefore AB: y = \frac{2p}{u+v}x + \frac{uv}{u+v}$

$\because F(\frac{p}{2}, 0)$

$\therefore uv = -p^2$

AB 中点 $(\frac{u^2+v^2}{4p}, \frac{u+v}{2})$

$\therefore \frac{u^2+v^2}{4p} = 3$

$u^2+v^2 = 12p$

$AB = AF+BF = \frac{u^2}{2p} + \frac{v^2}{2p} + p = 8$

$\frac{12p}{2p} + p = 8$

$p = 2$

$\therefore y^2 = 4x$

(2) $uv = -p^2 = -4$

$u^2+v^2 = 12p = 24$

$\therefore (u+v)^2 = u^2+v^2+2uv = 24-8 = 16$

$\therefore u+v = \pm 4$.

1° $u+v = 4$ 时.

$y = x-1$

2° $u+v = -4$ 时

$y = -x+1$

$\therefore y = x-1$ 或 $y = -x+1$

(1) 求椭圆 C 的方程；

(2) 过点 $(0, 2)$ 的直线 l 与 C 交于 A， B 两点．若 $OA \perp OB$（O 为坐标原点），求直线 l 的方程及弦长 $|AB|$。

$e = \frac{\sqrt{2}}{2} = \frac{c}{a}$

(1) $a = \sqrt{2}c$.

$\therefore b = c$

$\therefore \frac{x^2}{2c^2} + \frac{y^2}{c^2} = 1$

$\therefore \frac{2}{2c^2} + \frac{1}{c^2} = 1$

$c^2 = 2$

$\therefore a^2 = 2c^2 = 4$

$b^2 = c^2 = 2$

$\therefore \frac{x^2}{4} + \frac{y^2}{2} = 1$

(2) 设

1° 斜率不存在.

显然不符合题意舍去

2° 斜率存在

设 $y = kx + 2$.

$\therefore \begin{cases} \frac{x^2}{4} + \frac{y^2}{2} = 1 \\ y = kx + 2 \end{cases}$

$\therefore (1+2k^2)x^2 + 8kx + 4 = 0$.

$\Delta > 0$

设 $A(x_1, y_1)$

$B(x_2, y_2)$

$\therefore \begin{cases} x_1 + x_2 = \frac{-8k}{1+2k^2} \\ x_1 x_2 = \frac{4}{1+2k^2} \end{cases}$

$\because OA \perp OB$.

$\therefore \vec{OA} \cdot \vec{OB} = 0$.

$\therefore x_1 x_2 + y_1 y_2 = 0$.

$\therefore (1+k^2)x_1 x_2 + 2k(x_1 + x_2) + 4 = 0$.

$\therefore (1+k^2)\frac{4}{1+2k^2} + 2k \cdot \frac{-8k}{1+2k^2} + 4 = 0$

$\therefore 4(1+k^2) - 16k^2 + 4(1+2k^2) = 0$.

$\therefore k^2 = 2$

$\therefore k = \pm\sqrt{2}$

1° $k = \sqrt{2}$.

$\begin{cases} x_1 + x_2 = \frac{-8\sqrt{2}}{5} \\ x_1 x_2 = \frac{4}{5} \end{cases}$

$\therefore AB = \sqrt{k^2+1}\,|x_1 - x_2|$

$= \sqrt{2+1}\sqrt{(\frac{8\sqrt{2}}{5})^2 - 4 \times \frac{4}{5}}$

$= \frac{12}{5}$

2° $k = -\sqrt{2}$

$\begin{cases} x_1 + x_2 = \frac{8\sqrt{2}}{5} \\ x_1 x_2 = \frac{4}{5} \end{cases}$

$\therefore AB = \sqrt{k^2+1}\,|x_1 - x_2|$

$= \sqrt{2+1}\sqrt{(\frac{8\sqrt{2}}{5})^2 - 4 \times \frac{4}{5}}$

$= \frac{12}{5}$

$\therefore l_{AB}: y = \sqrt{2}x + 2$ 或 $y = -\sqrt{2}x + 2$.

$AB = \frac{12}{5}$

4. （17分） 已知等轴双曲线 $\Gamma: \frac{x^2}{a^2}-\frac{y^2}{b^2}=1\ (a>0,b>0)$ 的左、右焦点分别 F_1，F_2，且焦距为 $2\sqrt{2}$，A,B 分别是 Γ 在第二象限和第一象限上的一点，且 $AF_1 /\!/ BF_2$.

(1) 求 Γ 的方程；

(2) 若直线 AB 的斜率为 $\frac{1}{3}$，求直线 AF_1 的斜率；

(3) 若四边形 AF_1F_2B 的面积为 $2\sqrt{6}$，求直线 AF_1 的方程.



(1) $2c=2\sqrt{2}$

$c=\sqrt{2}$

$\because a=b$.

$\therefore c=\sqrt{a^2+b^2}=\sqrt{2}a$

$\therefore a=b=1$

$\therefore x^2-y^2=1$

(2) $F_1(-\sqrt{2},0)$

$F_2(\sqrt{2},0)$

1° AB 斜率不存在

舍去

2° AB 斜率存在

设 $y=kx+m\ (m<0)$

$\therefore \begin{cases} x^2-y^2=1 \\ y=kx+m \end{cases}$

$\therefore (1-k^2)x^2-2kmx-m^2-1=0$

$\therefore \begin{cases} 1-k^2\neq 0 \\ \Delta>0 \end{cases}$

设 $A(x_1,y_1)\quad B(x_2,y_2)$

$\therefore \begin{cases} x_1+x_2=\frac{2km}{1-k^2} \\ x_1x_2=-\frac{m^2+1}{1-k^2} \end{cases}$

$\because k=\frac{1}{3}$.

$\therefore \begin{cases} x_1+x_2=\frac{3}{4}m \\ x_1x_2=-\frac{9}{8}m^2-\frac{9}{8} \end{cases}$

$k=\frac{y_1}{x_1+\sqrt{2}}=\frac{y_2}{x_2-\sqrt{2}}$

$\therefore y_1x_2-\sqrt{2}y_1=y_2x_1+\sqrt{2}y_2$

$\therefore \frac{m}{\sqrt{2}}(x_2-x_1)=\frac{1}{3}(x_1+x_2)+2m$

$\frac{m}{\sqrt{2}}(x_2-x_1)=\frac{9}{4}m$

$x_2-x_1=\frac{9\sqrt{2}}{4}$

$\therefore (x_2-x_1)^2=(x_1+x_2)^2-4x_1x_2$

$\therefore \frac{81}{8}=\frac{9}{16}m^2+\frac{9}{2}m^2+\frac{9}{2}$

$\because m\geq 0$

$\therefore m=+\frac{\sqrt{10}}{3}$

$\therefore \begin{cases} x_1x_2=\frac{\sqrt{10}}{4} \\ x_1+x_2=-\frac{19}{8} \end{cases}$

$\therefore x_1=\frac{\sqrt{10}-9\sqrt{2}}{8}$

$y_1=\frac{3\sqrt{10}-3\sqrt{2}}{8}$

$\therefore k=\frac{\frac{3\sqrt{10}-3\sqrt{2}}{8}}{\frac{\sqrt{10}-\sqrt{2}}{8}}=3.$

(3) $S=\frac{1}{2}\cdot 2c\cdot(y_A+y_B)$

$2\sqrt{6}=\frac{1}{2}\cdot 2\sqrt{2}(y_A+y_B)$

$y_A+y_B=2\sqrt{3}$

$k(x_1+x_2)+2m=2\sqrt{3}$

$\frac{2k^2m}{1-k^2}+2m=2\sqrt{3}$

$m\cdot\frac{1}{1-k^2}=\sqrt{3}$

$m=\sqrt{3}(1-k^2)$

$k=\frac{y_1}{x_1+\sqrt{2}}=\frac{y_2}{x_2-\sqrt{2}}$

$\therefore m(x_2-x_1)=\frac{2\sqrt{2}m}{1-k^2}$

$x_2-x_1=\frac{2\sqrt{2}}{1-k^2}$

$\begin{cases} x_1+x_2=\frac{2km}{1-k^2} \\ x_2-x_1=\frac{2\sqrt{2}}{1-k^2} \end{cases}$

$\therefore \begin{cases} x_1=\frac{km-\sqrt{2}}{1-k^2} \\ x_2=\frac{km+\sqrt{2}}{1-k^2} \end{cases}$

$\therefore \frac{km-\sqrt{2}}{1-k^2}\cdot\frac{km+\sqrt{2}}{1-k^2}=-\frac{m^2+1}{1-k^2}$

$k^2m^2-2=-(m^2+1)(1-k^2)$

$k^2m^2-2=k^2m^2+k^2-m^2-1$

$k^2-3(1-k^2)^2+1=0.$

$\therefore 3k^4-7k^2+2=0.$

$\therefore k^2=2$ 或 $\frac{1}{3}$

1° $k^2=2$.

$m=-\sqrt{3}<0$ 舍

2° $k^2=\frac{1}{3}$

$m=2\sqrt{2}$

$y_1x_2-y_2x_1=\sqrt{2}(y_1+y_2)$

$\therefore x_1=\sqrt{3}k-\frac{3\sqrt{2}}{2}$

$y_1=\sqrt{3}-\frac{3\sqrt{2}}{2}k.$

1) $k=\frac{\sqrt{3}}{3}$

$\therefore A(1-\frac{3\sqrt{2}}{2},\frac{2\sqrt{3}-\sqrt{6}}{2})$

$k_{AF_1}=\frac{y_1}{x_1+\sqrt{2}}=\sqrt{3}$

2) $k=-\frac{\sqrt{3}}{3}$

$A(-1-\frac{3\sqrt{3}}{2},\frac{2\sqrt{3}+\sqrt{6}}{2})$

$k_{AF_1}=\frac{y_1}{x_1+\sqrt{2}}=-\sqrt{3}.$

$\therefore$ 得 $y=\sqrt{3}x+\sqrt{6}$

或 $y=-\sqrt{3}x-\sqrt{6}$